# MIT-ADT AI Grand Challenge 2026

## TITLE PAGE

- **Problem Statement ID** – Agri 3
- **Problem Statement Title** – Vision-Based Weed Detection with Targeted Spraying
- **Theme** – Agriculture
- **Category** – Hardware+Software
- **Team Name** – Conquerors

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# Proposed Solution

We propose a **AI-based robotic system** for smart agriculture system that uses a mobile phone camera and AI model to detect weeds in real-time.

The system:

- Captures images using mobile camera
- Detects weeds using AI model
- Sends signal via WiFi to Arduino
- Activates spray pump only on weeds

This ensures:

- Smart Farming
- Reduced Chemical Usage
- Low-Cost Implementation
- Uses Existing Smartphone (No Extra Camera Cost)

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# TECHNICAL APPROACH

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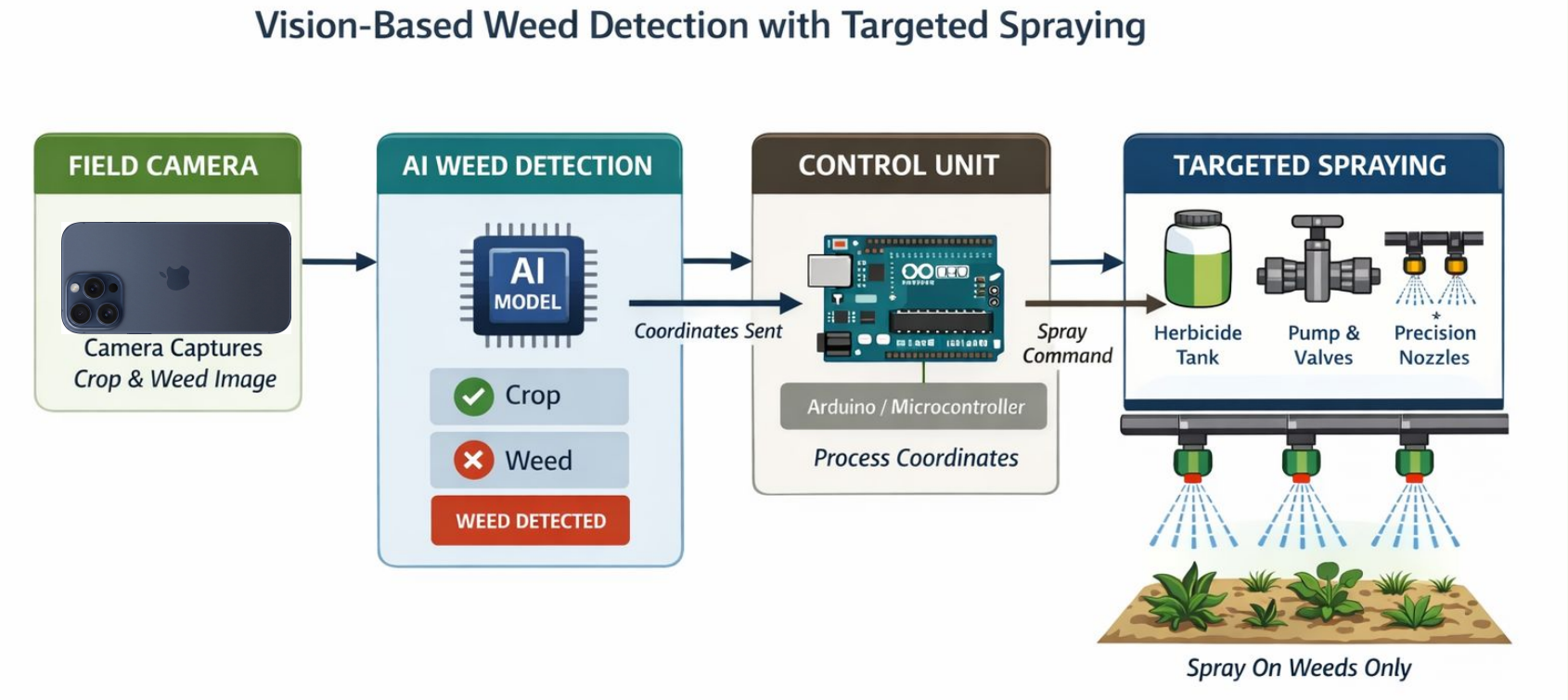
- Technologies/Methodology:



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# FEASIBILITY AND VIABILITY

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## Why This Works Practically

- YOLOv8n (nano) runs at ~15–25 FPS on mobile
- ESP32 latency < 100ms → near real-time actuation
- Spray precision achieved via grid-based calibration

## Core Technical Challenges

- Pixel-to-real world mapping
- Detection-to-spray delay
- Accuracy (false positives/negatives)
- Multiple weeds in one frame
- WiFi & hardware reliability

## Our Solution

1. Fixed Camera + Grid-Based Calibration
2. On-Device Processing with Minimal Data Transfer
3. Confidence Threshold + NMS Filtering
4. Centroid-Based Targeted Spraying
5. ESP32 with Lightweight WiFi Communication

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# IMPACT AND BENEFITS

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## 🌱 Environmental

→ Less pesticide use
→ Reduced soil & water pollution
→ Eco-friendly farming

## 🌾 Agricultural

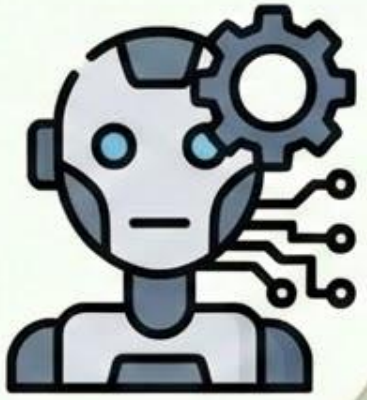
→ Better crop health & yield
→ Real-time weed detection
→ Precision spraying

## 💰 Economic

→ Low-cost (uses smartphone)
→ Saves labor & chemical cost
→ Higher farmer profit

## 🤖 Technology

→ AI & automation in farming
→ Promotes smart agriculture
→ Future-ready (drone upgrade)



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# RESEARCH AND REFERENCES

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- *Deep Learning–Based Approaches for Weed Detection in Crops* - Zhao H., Wang Y.
- *Detection of Commercial Crop Weeds using Machine Learning Algorithms* - Ramesh M. et al.
- *Hybrid Deep Learning Model for Weed Dataset Analysis* - Mishra S. et al.
- *Weed Detection Based on Deep Learning from UAV Imagery: A Review* - Sandoval-Pillajo C. et al.
- *Enhancing Weed Detection using Attention Mechanisms and Knowledge Distillation* - El Alaoui A., Mousannif H.